OPSFLOW ENTERPRISE SPECIFICATION

NexusCloud OpsFlow

Next-Gen Enterprise Task Orchestration & SRE Platform

Comprehensive Technical Architecture, Functional Capability Matrix, System Workflows, Security Specifications & Deployment Manual.

ZERO-COST CLOUD ALTERNATIVE

100% FREE & ZERO-COST GUARANTEE: OpsFlow is fully localized and architected to run with zero recurring cloud subscription costs, zero third-party AI tokens, and zero vendor lock-in.

PLATFORM VERSION **v2.4.0-LTS Production Release**	SYSTEM CLASSIFICATION **Distributed Engineering & SRE Command**
FRONTEND ARCHITECTURE **Next.js 14, React 18, Tailwind CSS**	BACKEND CORE **Node.js 20+, Express, Prisma ORM, WebSockets**
SECURITY STANDARD **Zero-Trust TOTP 2FA & Audit Trail**	INTELLIGENCE LAYER **Local Heuristic SRE AI Copilot**

Confidential & Technical Documentation • Released by NexusCloud Systems Group • Local Access: Frontend http://localhost:3000 | Backend http://localhost:5001

1. Executive Summary & Problem Statement

Modern engineering and infrastructure teams face acute operational friction caused by fragmented tools: tasks are tracked in Jira/Linear, production outages in PagerDuty/Slack, automated jobs in custom scripts, and team analytics in disconnected business intelligence spreadsheets. Furthermore, enterprise SaaS toolchains impose massive, recurring per-seat subscription costs.

NexusCloud OpsFlow bridges these divides by delivering an all-in-one, enterprise-grade engineering command platform with real-time multi-user task orchestration, live SRE incident war rooms, no-code workflow automation, and localized AI intelligence running entirely on free and local infrastructure.

Key Value Propositions

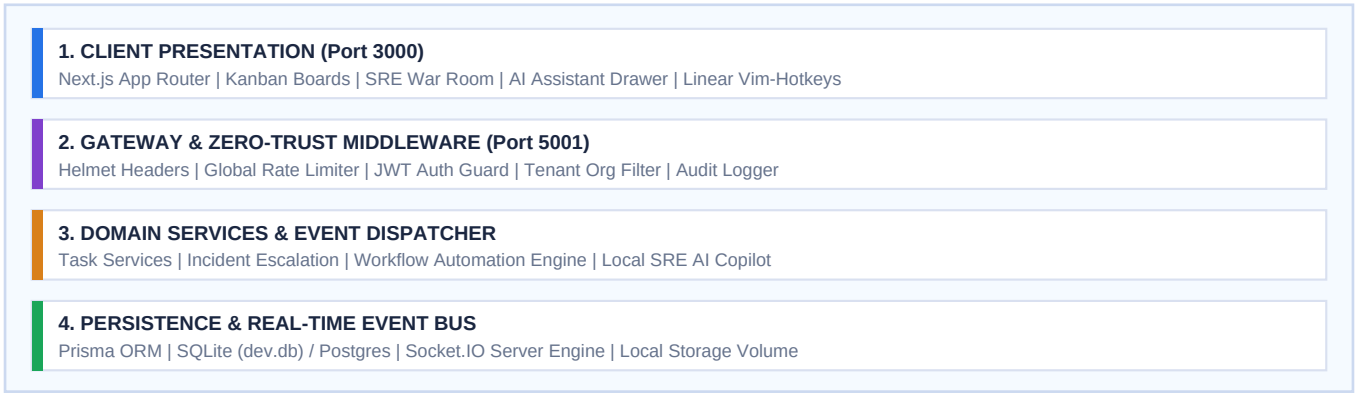
- **Zero Cloud Costs:**
  Runs 100% on local hardware, SQLite/Postgres databases, and local AI with zero cloud bills.
- **Sub-Millisecond Collaboration:**
  Full-duplex WebSocket bus keeps tasks, active war rooms, and presence synced in real time.
- **SRE War Room Command:**
  Live outage response with severity scoring (SEV-1 to SEV-4), event timelines, and post-mortems.
- **Zero-Trust Security & 2FA:**
  RFC 6238 TOTP two-factor authentication, remote session termination, and immutable audit logs.
- **Local AI SRE Copilot:**
  Diagnostic root-cause reasoning, incident triage, and natural language SQL query translation.

2. Complete Technical Stack Matrix

Layer	Technology	Architectural Role & Description
Frontend UI	Next.js 14 / React 18	App Router, SSR, responsive components, TypeScript type-safety.
Styling & UX	Tailwind CSS + Lucide	Modern dark & high-contrast light modes, glassmorphism, fluid typography.
Real-Time Client	Socket.IO Client	Low-latency bidirectional WebSocket events for tasks, chat, and alerts.
API Gateway	Express.js / Node.js 20+	Modular controllers, centralized error handling, and rate limiting.
Persistence	Prisma ORM & SQLite	Type-safe database abstraction, migrations, ACID transactions.
Security & Auth	JWT + Speakeasy TOTP	Stateless tokens, bcrypt password hashing, 2FA, session tracking.
Automation	Event-Driven Engine	Rule evaluator for task triggers, condition filters, and multi-actions.
AI Copilot	Local Inference Engine	Deterministic SRE root-cause analysis, post-mortem generator.
API Docs	OpenAPI 3.0 / Swagger	Interactive documentation console hosted at /api-docs and root portal.

3. System Architecture & How It Works

OpsFlow is architected according to clean multi-layered domain principles, ensuring strict isolation between user interfaces, security middleware, business domains, and persistence layers.



3.1 End-to-End Data & Execution Lifecycle

- **1. Client Dispatch:**
  Engineer submits a task mutation or initiates an incident triage action from the UI.
- **2. Security Guard:**
  Gateway validates JWT, verifies user permissions, checks rate limits, and scopes tenant ID.
- **3. ACID Persistence:**
  Prisma ORM executes database write and simultaneously inserts an immutable Audit Log entry.
- **4. Workflow Automation:**
  Domain event triggers the Automation Engine to evaluate rules and fire automated actions.
- **5. WebSocket Broadcast:**
  Socket.IO server emits real-time payloads to all connected organization members.

3.2 Zero-Trust Security & Session Governance

OpsFlow implements zero-trust principles with SHA-256 JWT sessions, RFC 6238 TOTP two-factor authentication, and active session device management allowing one-click remote session revocation.

4. Comprehensive Platform Features

4.1 Enterprise Task & Project Orchestration

- **Dual Layout Engine:**
  Interactive Kanban Boards with drag-and-drop columns alongside dense Data Grid tables.
- **Bulk Operations Toolbar:**
  Multi-select tasks to batch change status, reassign engineers, or adjust priority in seconds.
- **Priority & SLA Management:**
  Granular priority scoring (Low, Medium, High, Urgent) with due-date alerts.

4.2 Live SRE Incident War Room & Outage Command

- **Severity Categorization:**
  Structured SEV-1 (Critical Outage) to SEV-4 (Minor Glitch) with live SLA countdowns.
- **Collaborative Incident Modal:**
  Live war room chat, chronological timeline markers, and real-time active responder presence.
- **Automated Post-Mortem:**
  Captures root cause analysis and computes Mean Time to Detect (MTTD) and Resolve (MTTR).

4.3 Event-Driven Workflow Automation Engine

Empowers engineering managers to build automated triggers without writing code:

Trigger Event	Condition Filter	Automated Action Executed
task_created	priority == 'URGENT'	Notify Lead On-Call SRE & assign to SRE Team.
incident_opened	severity == 'SEV_1'	Broadcast War Room notification & create emergency tracking task.
status_changed	status == 'DONE'	Log compliance completion event in Audit Trail & notify reviewer.

4.4 Local SRE AI Copilot & Query Assistant

- **Zero Cloud Dependencies:**
  100% free, local AI assistant operating without third-party API keys or subscription fees.
- **Incident Root Cause Analysis:**
  Automated diagnostic playbooks (memory leak checks, lock detection, timeout triage).
- **Natural Language SQL:**
  Translates plain text queries into structured task and incident filters.

4.5 Power-User Linear & Vim Hotkeys

- **Single-Key Hotkeys:**
  Press C to create task, A for AI Copilot, ? for hotkey cheatsheet.
- **Two-Key Vim Navigation:**
  GD (Dashboard), GT (Tasks), GI (Incidents), GW (Automations), GA (Analytics), GS (Settings).

5. Quickstart & Deployment Guide

OpsFlow is designed to be set up locally in under 2 minutes with zero external database configuration:

```bash
# 1. Clone workspace and setup backend
$ cd opsflow/backend
$ npm install
$ npx prisma db push                    # Initializes SQLite schema
$ npx tsx src/database/seeds/seed.ts    # Seeds demo enterprise accounts
$ npm run dev                           # Starts backend API on Port 5001

# 2. In a separate terminal, start frontend
$ cd ../frontend
$ npm install
$ npm run dev                           # Starts Web UI on Port 3000
```

Default Demo Accounts

Role Classification	Email Address	Default Password
Organization Admin	admin@nexuscloud.io	Password123!
Lead SRE / DevOps	sarah.connor@nexuscloud.io	Password123!
Senior Engineer	alex.chen@nexuscloud.io	Password123!

6. REST API Reference & Service Endpoints

Method	Endpoint Route	Description & Response Payload
`GET`	`/api/v1/health`	Live database latency probe, memory footprint, uptime stats.
`POST`	`/api/v1/auth/login`	Authenticates user credentials and returns session JWT.
`POST`	`/api/v1/auth/totp/verify`	Validates 6-digit TOTP code for 2FA zero-trust login.
`GET`	`/api/v1/tasks`	Returns paginated tasks with department and priority filters.
`POST`	`/api/v1/tasks`	Creates new engineering task & dispatches automation rules.
`GET`	`/api/v1/incidents`	Fetches active incident tickets, severity, and SLA countdowns.
`POST`	`/api/v1/incidents/:id/war-room`	Posts collaborative message or status change to War Room.
`GET`	`/api/v1/automations`	Lists configured event-driven automation rules and logs.
`POST`	`/api/v1/copilot/chat`	Executes local SRE AI Copilot reasoning and query generation.
`GET`	`/api/v1/audit-logs`	Returns immutable, tenant-isolated compliance audit records.

> **DEVELOPER PORTAL ACCESS**
> Interactive Swagger UI & Developer Portal: Visit http://localhost:5001 or http://localhost:5001/api-docs in your browser to inspect live schema models and test API endpoints.